

# EMo Learners | Solved Model Exam Paper

## DATABASE MANAGEMENT SYSTEM (DBMS) MODEL EXAM

### SISTec Academic Mid-Semester Pattern — Semester IV

#### Unit 1 (Introduction to DBMS) & Unit 2 (Relational Database Model)

Subject: Database Management System (CSIT-405)

Total Solved Marks: 28 Marks

Layout Structure: Part A (MCQ), Part B (Short), Part C (Long Choice)

Branding Highlight: EMO Learners Exclusive  
Solved Paper

## PART A: OBJECTIVE MULTIPLE CHOICE QUESTIONS (6 MARKS)

Instructions: Answer all twelve questions. Each question carries exactly 0.5 marks.

**Q.1 Which anomaly is a direct consequence of storing duplicate data records across multiple isolated files in a file system?**

- A. Data Isolation
- B. Data Redundancy and Inconsistency
- C. Transaction Abort
- D. Lossless Join Constraint

**Answer: B. Data Redundancy and Inconsistency**

Explanation: Data redundancy occurs when the same information is duplicated in several files. When one copy is updated and others are not, it leads to data inconsistency.

**Q.2 Which schema represents the actual physical layout of database files on secondary disk storage drives?**

- A. External Schema
- B. Conceptual Schema
- C. Internal (Physical) Schema
- D. Logical Abstraction Level

**Answer: C. Internal (Physical) Schema**

Explanation: The internal schema describes the low-level physical storage structure, database index paths, block allocations, and file access layouts.

**Q.3 The database administrator's ability to modify logical table structures without changing user query scripts is called:**

- A. Physical Data Independence
- B. Logical Data Independence
- C. Referential Integrity
- D. Trivial Functional Dependency

**Answer: B. Logical Data Independence**

Explanation: Logical data independence is the ability to change the conceptual logical schema without affecting the external schemas or application view interfaces.

**Q.4 In the Entity-Relationship (ER) model, a weak entity set is represented visually by which diagram notation?**

- A. Dashed Oval
- B. Double Rectangle
- C. Double Diamond
- D. Thick Line

**Answer: B. Double Rectangle**

Explanation: A weak entity set lacks a primary key and relies on a strong identifying parent entity. In ER diagrams, it is drawn as a double rectangle.

**Q.5 Which database key is defined as a minimal set of attributes that uniquely identifies a row in a relation table?**

- A. Super Key
- B. Candidate Key
- C. Foreign Key
- D. Composite Attribute Key

**Answer: B. Candidate Key**

Explanation: A candidate key is a minimal superkey, meaning no proper subset of its attributes can uniquely identify tuples in the relation.

**Q.6 Which relational algebra fundamental operation is used to extract specific columns from a relation table?**

- A. Selection
- B. Projection
- C. Rename
- D. Natural Join

**Answer: B. Projection**

Explanation: Projection is a unary fundamental operation denoted by the Greek letter pi that filters columns vertically, discarding the rest.

**Q.7 The integrity rule stating that no primary key attribute column can contain a NULL value is called:**

- A. Referential Integrity
- B. Entity Integrity
- C. Domain Constraint
- D. Guaranteed Access Rule

**Answer: B. Entity Integrity**

Explanation: Entity integrity guarantees that each row remains uniquely identifiable. Since primary keys serve this purpose, they can never contain a null value.

**Q.8 In a Hospital ER diagram, a doctor treats multiple patients, and a patient is treated by multiple doctors. This mapping is:**

- A. One-to-One (1:1)
- B. One-to-Many (1:N)
- C. Many-to-Many (M:N)
- D. Ternary Relationship

**Answer: C. Many-to-Many (M:N)**

Explanation: A doctor sees many patients, and a patient can consult many doctors. Therefore, the relationship treats is a many-to-many cardinality mapping.

**Q.9 Which low-level file system operation is responsible for allocating memory buffers and establishing an active OS handle to read files?**

- A. Read
- B. Open
- C. Flush
- D. Close

**Answer: B. Open**

Explanation: The open operation locates the file on disk, validates permissions, allocates memory buffers, and establishes a logical link for subsequent I/O.

**Q.10 What is the mathematical definition of a database table row in formal relational database theory?**

- A. Tuple
- B. Domain
- C. Relation Schema
- D. Intension

**Answer: A. Tuple**

Explanation: In relational theory, a table is a relation, a column is an attribute, and a row is formally referred to as a tuple.

**Q.11 The top-down abstraction process of breaking down a high-level entity into specialized sub-entities is called:**

- A. Generalization
- B. Specialization
- C. Aggregation
- D. Entity Mapping

**Answer: B. Specialization**

Explanation: Specialization is a top-down database design process that splits a generic entity into specialized sub-entities based on distinct attributes.

**Q.12 Which relational algebra operation combines rows from two tables by automatically matching and filtering on their shared column domains?**

- A. Cartesian Product
- B. Natural Join
- C. Left Outer Join
- D. Set Difference

**Answer: B. Natural Join**

Explanation: A natural join combines rows from two relations R and S by matching values across their common attributes, automatically discarding duplicate join columns.

**PART B: SHORT ANSWER QUESTIONS (8 MARKS)****Section 1: Unit 1 (Introduction to DBMS)**

Instructions: Answer any two of the following three questions. Each question carries 2 marks.

**Q.1 Differentiate between logical data independence and physical data independence.**

[Marks: June 2022]

**Answer:**

Traditional file systems and Database Management Systems differ in how they achieve logical and physical abstraction layers:

1. Logical Data Independence: The ability to modify the conceptual logical schema (such as adding a table column or changing database design rules) without needing to alter higher-level external schemas, user views, or query scripts.
2. Physical Data Independence: The ability to modify the internal physical schema (such as switching from HDDs to SSDs, rebuilding B-Tree indexes, or changing data file storage paths) without requiring any changes to conceptual logical schemas or application codes.

**Q.2 Describe various disadvantages of conventional file systems compared to a Database Management System.**

[Marks: May 2018, Dec 2020, June 2022]

**Answer:**

A conventional file based system stores data in isolated files using operating system structures, which introduces several limitations:

1. Data Redundancy and Inconsistency: Because files are isolated, identical data is duplicated across multiple locations. If one file is updated while copies are not, the database enters an inconsistent state.
2. Difficulty in Accessing Data: Searching for records requires writing customized, complex programming scripts (e.g. in C or Java) for every unique search request, as there is no query language like SQL.
3. Lack of Security and Concurrency Control: Conventional file systems do not have security checks to limit access permissions for individual users, nor do they support simultaneous read/write locks, causing file corruption during concurrent operations.

**Q.3 Explain the different categories of database users with their operational roles.**

[Marks: May 2018]

**Answer:**

Database users represent different groups of people who interact with the system based on their operational roles:

1. Application Programmers: Software developers who design and write database interface applications using languages like C++ or Java and database APIs (e.g. JDBC, ODBC) to help parametric users perform transactions.
2. Sophisticated (Analyst) Users: Data scientists and analysts who write custom, complex query scripts (such as SQL) directly to the database engine to extract summaries for business decisions.
3. Naive (Parametric) Users: Front-end operators (e.g. bank tellers, airline ticketing clerks) who run database operations through simple pre-built application forms, completely unaware of the conceptual schema.

## Section 2: Unit 2 (Relational Model & ER)

Instructions: Answer any two of the following three questions. Each question carries 2 marks.

### Q.4 Differentiate Specialization and Generalization with the help of examples.

[Marks: June 2020, June 2022]

#### Answer:

Generalization and Specialization are advanced conceptual design techniques used to construct superclasses and subclasses in extended ER models:

1. Specialization (Top-Down): The process of breaking down a generic high-level entity into multiple lower-level sub-entities based on distinct attributes. For example, a generic entity Employee is specialized down into Salaried\_Staff (monthly pay) and Hourly\_Contractors (hourly pay) sub-entities.
2. Generalization (Bottom-Up): The process of merging multiple distinct lower-level entities that share common features into a single, unified high-level super-entity. For example, distinct entities Car, Truck, and Motorcycle are generalized into a high-level super-entity Vehicle because they all share common attributes like Model and Fuel\_Type.

### Q.5 Write a short note on strong and weak entity sets.

[Marks: Nov 2019]

#### Answer:

Logical database modeling classifies entities into two categories based on key attributes and dependency rules:

1. Strong Entity Set: A strong entity set contains a primary key attribute that uniquely identifies each entity row inside the set. It stands independently without requiring relationships with other tables. It is drawn as a standard single rectangle box. An example is the Employee table with the primary key Emp\_ID.
2. Weak Entity Set: A weak entity set does not have a primary key attribute and cannot be uniquely identified on its own. It depends on an identifying relationship with a strong parent entity to exist. It uses a partial key (discriminator) drawn with a dashed underline to identify rows under a parent key. It is drawn as a double rectangle box, and its identifying relationship is drawn as a double diamond. An example is a Dependent of an Employee.

**Q.6 Explain Relational Schema Intension and Extension in detail.****Answer:**

In the relational database model, a relation contains two distinct structural characteristics:

1. **Relational Intension (Schema):** The permanent structural definition of the relation table. It specifies the table name, the names of all columns (attributes), and their permitted data types (domains). It is static, defined using DDL commands, and rarely changes. For example, `Student(Roll: INT, Name: VARCHAR, Age: INT)` is the intension.
2. **Relational Extension (State/Instance):** The dynamic collection of rows (tuples) present in the table at a specific point in time. It represents the actual data values stored in the table cells. It is dynamic, altered continuously using DML commands (insert, delete, update). For example, the row `(101, 'John', 21)` represents a single tuple in the extension.

## PART C: LONG ANSWER & TRACING QUESTIONS (14 MARKS)

### Set 1: Unit 1 (Introduction to DBMS)

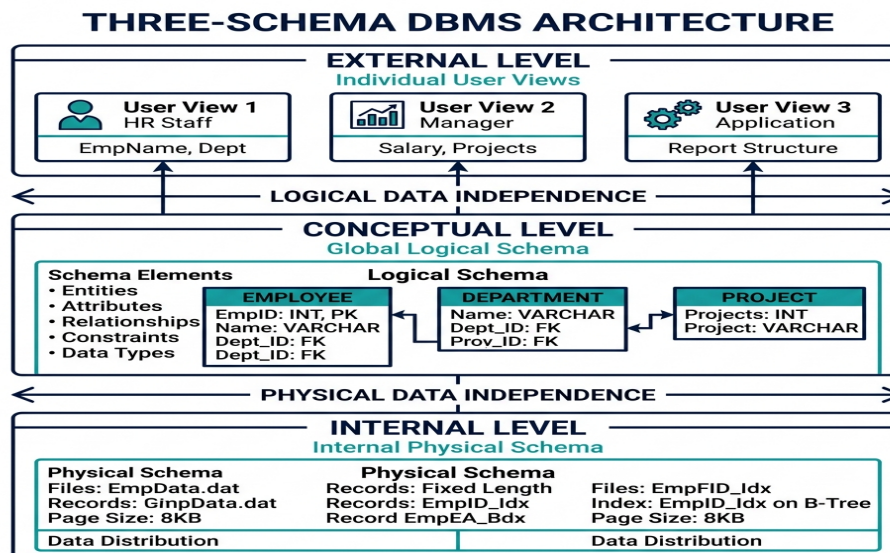
Instructions: Answer any one of the following two questions (with internal choice). The question carries 7 marks.

**Q.7 [a]** Explain the Three-Schema Architecture of a database system in detail, detailing the role of physical, conceptual, and external schemas.

[Marks: Dec 2017, June 2022]

**Answer:**

The Three-Schema Architecture (also known as the ANSI-SPARC architecture) divides the database into three abstraction layers to hide hardware details from users, making it easy to change one layer without breaking the others.



### 1. The Three Abstraction Levels:

- **External Level (View Schema):** Describes only the part of the database a specific user group cares about (e.g. a student sees grades, while an accounts officer sees fee balances). It consists of multiple external views.
- **Conceptual Level (Logical Schema):** Describes what data is stored and what relationships exist. It defines tables, domains, primary keys, and logical rules, completely hiding physical storage designs.
- **Internal Level (Physical Schema):** Describes how data is physically stored in secondary storage devices (hard disks, SSDs). It details record formats, block sizes, indexes, and compression schemes.

### 2. Data Independence Mapping:

Data independence refers to the ability to modify schema definitions at one level without forcing modifications at higher levels:

- Logical Data Independence: The ability to modify the conceptual schema (e.g. adding a new table column or renaming an entity) without needing to change existing external view programs or SQL queries.
- Physical Data Independence: The ability to modify the internal physical schema (e.g. switching storage from HDD to SSD, adding B-Tree indexes, or renaming data directories) without needing to change conceptual logic schemas or user applications.

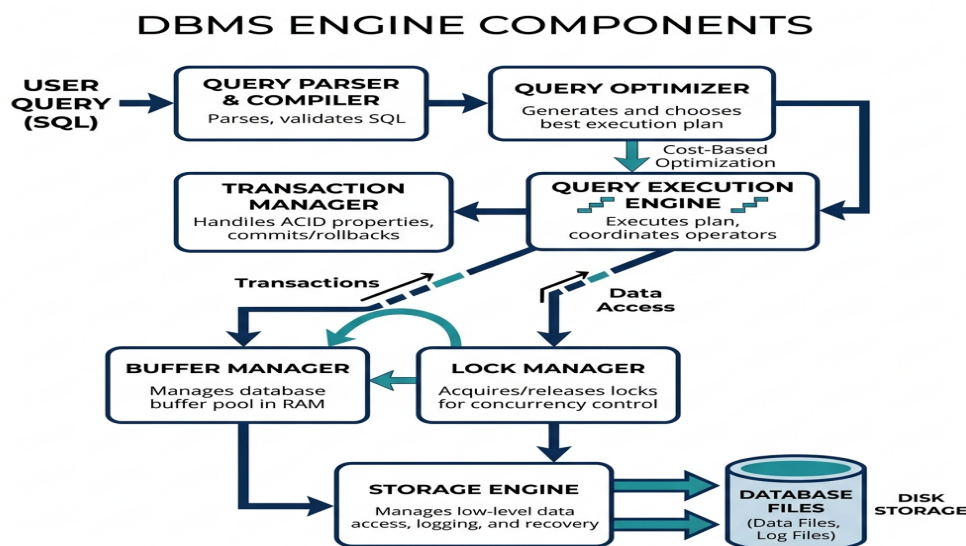
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**Q.7 [b] Define Database Management System. Draw and explain the complete functional architecture of DBMS engine components.**

[Marks: Nov 2019, Dec 2020, June 2022]

**Answer:**

A Database Management System (DBMS) consists of multiple coordinated software components that handle query processing, transactional safety, buffer caching, and physical storage interface.



**Functional Components of DBMS Architecture:**

- Query Parser and Compiler: This subsystem acts as the entry point for all incoming database requests. It performs lexical and syntactic analysis on raw SQL text, checking it against the system catalog metadata to verify table names, column existence, and user privileges. The parsed query is then translated into a relational algebra expression tree.
- Query Optimizer: The query optimizer is the brain of the compiler. It analyzes multiple equivalent relational algebra trees and evaluates execution paths using a cost-based model. It calculates CPU and disk I/O costs based on database statistics (such as table sizes and index distributions) to select the most efficient plan.
- Query Execution Engine: The execution engine takes the compiled execution plan and coordinates physical operations. It makes calls to the buffer manager and transaction engine to fetch, join, sort, and write database blocks.



- Buffer Manager: Responsible for managing the database's RAM cache. It decides which pages of data should be loaded from disk into memory, and which pages should be evicted back to disk using caching algorithms like Least Recently Used (LRU).
- Transaction and Lock Manager: Monitors active database sessions to guarantee the ACID properties. It manages row-level locks to prevent dirty reads or lost updates and writes recovery logs to disk to handle system rollbacks.

**Set 2: Unit 2 (Relational Model & ER)**

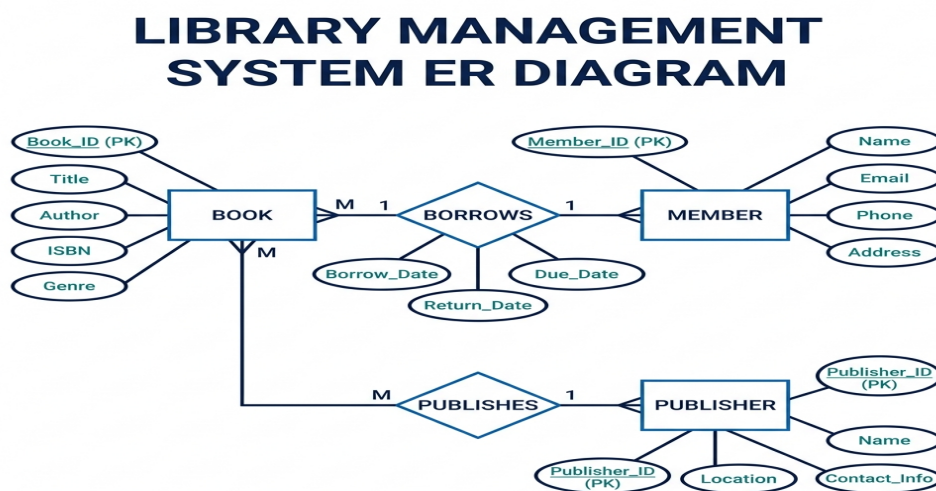
Instructions: Answer any one of the following two questions (with internal choice). The question carries 7 marks.

**Q.8 [a] Define ER diagram. Draw an ER diagram for a Library Management System, showing all entities and relationships.**

[Marks: May 2018, Dec 2020]

**Answer:**

An Entity-Relationship (ER) diagram represents the database schema visually using rectangles (entities), circles (attributes), and diamonds (relationships).

**Library ER Schema Elements:**

- Entities & Attributes: Book (Book\_ID, Title, Author, Price), Member (Member\_ID, Name, Email, Phone), and Publisher (Publisher\_ID, Name, City).
- Relationships & Cardinality:
  1. Borrows (Many-to-Many): Multiple members can borrow multiple books. Cardinality is M:N.
  2. Publishes (One-to-Many): One publisher publishes multiple books, but a book belongs to only one publisher. Cardinality is 1:N.

----- OR -----

**Q.8 [b] Explain the fundamental operations of Relational Algebra (Selection, Projection, Union, Set Difference, Cartesian Product, and Rename) with examples.**

[Marks: June 2017, May 2018, Dec 2020]

**Answer:**

Relational Algebra is a formal procedural query language that takes one or more relations as input and produces a new relation as output. Its six fundamental operations are explained below in descriptive textbook form:

1. Selection ( $\sigma$ ): A unary operator that filters rows based on a boolean criteria, extracting subset rows from a table. For example,  $\sigma_{\text{Age} > 20}(\text{Students})$  filters student rows where age exceeds 20.
2. Projection ( $\pi$ ): A unary operator that selects specific attribute columns, discarding the rest and removing duplicates from the result set. For example,  $\pi_{\text{Name, Roll}}(\text{Students})$  extracts only the Name and Roll columns.
3. Union ( $\sqcup$ ): Combines rows from two relations. The relations must be union-compatible (have same number of columns with matching domain types). Discards duplicates.
4. Set Difference ( $-$ ): Finds rows that are present in the first relation R but absent in the second relation S (R and S must be union-compatible).
5. Cartesian Product ( $\times$ ): Combines every row of relation R with every row of relation S, producing all possible row combinations. If R has m rows and S has n rows, the product has m times n rows.
6. Rename ( $\rho$ ): Alters the logical attribute names or the table identifier name.